CCC-C-437D
June 13, 1973
SUPERSEDING
Fed. Spec. CCC-C-437C
March 17, 1969

FEDERAL SPECIFICATION

CLOTH, BROADCLOTH, COTTON MERCERIZED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers cotton broadcloth.

1.2 Classification.

1.2.1 Types and classes. The cloth shall be of the following types and classes, as specified (see 6.2 and 6.4).

Type I	-	148 by 72, 3.5 ounces per square yard.
Type II	-	140 by 58, 3.2 ounces per square yard.
Type III	-	98 by 56, 3.0 ounces per square yard.
Class 1	-	Bleached.
Class 2	-	Unbleached.
Class 3	-	Dyed.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

V-T-276	-	Thread, Cotton.
PPP-P-1134	-	Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics, (Excluding Duck Fabrics).

Federal Standards:

Fed. Std. No. 4	-	Glossary of Fabric Imperfections.
Fed. Std. No. 191	-	Textile Test Methods.

FSC 8305

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(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers, at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Ft. Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

PUBLICATION

Rules and Regulations under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

3. REQUIREMENTS

3.1 Standard sample. When a standard sample is available, the bleached or dyed cloth shall match the sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material.

3.2.1 Yarn. The yarn shall be made from cotton which has been carded, or combed as specified in table I, drawn and spun into single or ply yarn.

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3.3 Color.

3.3.1 Dyed cloth (class 3). The color shall be as specified (see 6.2). The use of dyes and substances containing elementary sulfur compounds capable of oxidation to sulfuric acid is prohibited. The dyestuff shall be chosen and applied so that the dyed and finished cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.4. When a standard sample is not available the dyed and finished cloth shall show not more than a slight trace of labile sulfur as defined in the test method, when tested as specified in 4.4.

3.3.2 Bleached cloth (class 1). When white is specified, the cloth shall be chemically bleached white and may be supplemented with optical brighteners which fluoresce to the blue region. The degree of whiteness shall match the standard sample. The white finished bleached cloth shall not discolor to a degree greater than that shown by the standard sample when tested as specified in 4.4. When no standard sample is available, the discoloration shall be not less than "fair" when tested as specified in 4.4.

3.3.3 Matching. The color shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.3.4 Colorfastness. The dyed and finished (class 3 and flame-resistant treated) cloth shall show colorfastness to laundering (after 3 cycles), perspiration, crocking and light equal to or better than the standard sample. When a standard sample is not available or not designated as applicable to colorfastness, the dyed cloth shall show "good" colorfastness to laundering (after 3 cycles), perspiration, light and no more crocking than Munsell value 8.5 when tested as specified in 4.4.

3.4 Physical requirements. The finished cloth, except when flame-resistant treated cloth is required, shall conform to the requirements shown in table I when tested as specified in 4.4.

3.4.1 Width. The width shall be as specified and shall be the minimum acceptable width inclusive of selvages (see 6.2).

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TABLE I. Physical requirements

Type	Yarn	Weight per sq. yd. (min) ounces	Yarns per inch (min) Warp	Yarns per inch (min) Filling	Breaking strength (min) Pounds Warp	Breaking strength (min) Pounds Filling	Weave
I	Combed	3.5	148	72	68	32	Plain
II	Combed	3.2	140	58	70	25	Plain
III	Carded	3.0	98	56	55	24	Plain

3.5 Finish. The cloth shall be singed, desized, (when required), and mercerized.

3.6 Nonfibrous materials. The starch and protein content including chloroform-soluble and water-soluble material shall not exceed 2.0 percent when tested as specified in 4.4.

3.7 pH. The pH value of the water extract of the finished cloth shall be no less than 5.0 nor more than 8.5 when tested as specified in 4.4.

3.8 Shrinkage. The cloth shall be preshrunk to a residual shrinkage of not more than 1.0 percent in either the direction of the warp or of the filling when tested as specified in 4.4. The preshrinking process used shall not be identified on the cloth, the ticket, or on the package.

3.9 Seam efficiency. The finished cloth shall have a seam efficiency of not less than 85 percent when tested as specified in 4.4.

3.10 Put-up. Unless otherwise specified (see 6.2), the finished cloth shall be furnished in continuous lengths each not less than 50 yards. Each length shall be put-up in flat folded pieces as specified in PPP-P-1134.

3.11 Fiber identification. Each piece shall be labeled, ticketed or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.12 Flame resistance. When required, the cloth shall be treated with a flame-resistant compound. The finished treated cloth shall have an average time of flaming of not more than 2.0 seconds and an average length of char of not more than 5.0 inches, both initially and following 15 launderings, when tested as specified in 4.4 (see 6.2 and 6.5).

3.12.1 Except as specified below, flame-resistant treated cloth shall meet all requirements, herein, of the untreated cloth:

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(a) Breaking strength. The breaking strength of the warp and filling shall be not less than 75 percent of established strength values of the untreated cloth.

(b) Tearing strength. The tearing strength of the treated cloth shall be not less than 50 percent of that of the untreated cloth.

(c) Yarns per inch (filling). The yarns per inch in the filling direction of the treated cloth shall be not less than 90 percent of the amount found in the untreated cloth.

(d) Weight. The weight of the treated cloth shall not exceed 130 percent of the weight of the untreated cloth.

(e) Finish. The treated cloth shall show no appreciable change in stiffness or hard than that of the untreated cloth, and shall not exhibit powdering of the flame-resistant compound.

3.12.2 Toxicity. The treated cloth shall be noninjurious and shall not produce skin irritation to personnel under normal conditions of its end use. Formula approval of the flame-resistant compound shall be obtained by the contractor from the procuring activity prior to production.

3.13 Marking. The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word "Face" at each end of the piece. In addition, flame-resistant treated cloth shall be identified by a legible ink stamping of the words "Flame-resistant".

3.14 Workmanship. The finished cloth shall conform to the quality and grade of product established by this specification. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

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4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 Examination of the end item. Examination of the end item shall be in accordance with 4.2.2.1 thru 4.2.2.4.

4.2.2.1 Yard-by-yard examination. Each piece in the sample shall be examined on the face side only. When the total yardage in the piece does not exceed 100 yards, the entire yardage in the piece shall be examined. When the total yardage in the piece exceeds 100 yards, only 100 yards shall be examined. All defects as defined in section I of Fed. Std. No. 4, which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.2.1.1 except that only those slubs and knots which exceed the limits shown on Sears Fabric Defect Scales (see 6.6), "D" for slubs and "C" for knots, shall be scored. No linear yard shall be penalized more than 4 points. The sample size shall be 20 pieces selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceed the following values:

32.0 points for white cloth.
28.0 points for dyed cloth.

For determination of acceptability on the individual piece quality basis, the lot shall be unacceptable if two or more individual pieces exceed the following values:

48.0 points for white cloth.
42.0 points for dyed cloth.

If one piece exceeds the point level, a second sample of 20 pieces shall be selected for individual piece quality examination only. The lot shall be unacceptable if one or more pieces in the second sample exceed the point level. Point computation for lot quality and individual piece quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{total yards inspected}} = \text{Points per 100 square yards}$$

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4.2.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension	-	one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	-	two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	-	three points
For defects exceeding 9 inches in any dimension	-	four points

The following defects, when present, shall be scored four points for each yard in which they occur:

Baggy, ridgy or wavy cloth.
Objectionable odor.
Width less than specified.
Poor dye penetration, mottled, streaky or cloudy.

4.2.2.2 Examination for length.

4.2.2.2.1 Individual pieces. During the yard-by-yard examination, each piece in the sample shall be examined for length. Any length found to be less than the minimum specified, or more than two yards less than the length marked on the ticket, shall be considered a defect with respect to length. The lot shall be uancceptable if two or more pieces in the sample are defective in respect to length.

4.2.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of pieces in the sample is less than the total of the lengths marked on the tickets.

4.2.2.3 Examination for shade. During the yard-by-yard examination, each piece in the sample shall be examined for shade. Any piece in the sample, off-shade, shaded side to side, side to center, or end to end, shall be cause for rejection of the entire lot represented by the sample.

4.2.2.4 Examination for face markings, identification of preshrinkage process and non-compliance with the Textile Fiber Products Identification Act. During the yard-by-yard examination, each piece in the sample shall be examined for these defects. The lot shall be unacceptable if two or more pieces in the sample contain identification of the preshrinkage process by name or trademark on the cloth or ticket, or are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act,

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or the face stamping missing from either, or both ends, or the face stamping on the wrong side. In addition to the above listed defects, examination shall be made to determine whether flame-resistant finished cloth has been properly identified on each piece.

4.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1134 to determine that packaging, packing and marking comply with section 5 requirements.

4.4 Testing of the end item. The methods of testing specified in Fed. Std. No. 191, wherever applicable, and as listed in table II shall be followed. The physical and chemical values specified in section 3, except where otherwise specified, apply to the results of the determinations made on the sample unit for test purposes as specified in the applicable test method. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be 1/2 yard of the untreated cloth and 4-1/2 continuous yards full width of the finished cloth. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. The sample size shall be as follows:

Lot size (yards)	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE II. Test methods

Characteristic	Requirement paragraph	Test method
Identification of cotton	3.2	1200 1/
Combed yarns - types I and II	3.2.1	1/
Presence of labile sulfur	3.3.1	2020
Discoloration (bleached, class 1)	3.3.2	5660 2/ 3/
Colorfastness to:		
Laundering (after 3 cycles)	3.3.4	5610 4/ 5/
Perspiration	3.3.4	5680
Crocking	3.3.4	5651
Light	3.3.4	5660
Weight	3.4	5041

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TABLE II. Test methods (cont'd)

Characteristic	Requirement paragraph	Test method
Yarns per inch:		
Warp	3.4	5050
Filling	3.4	5050
Breaking strength:		
Warp	3.4	5100
Filling	3.4	5100
Weave	3.4	Visual 2/
Mercerization	3.5	Microscopic examination for swelling of cotton fibers 1/ 2/
Singeing	3.5	1/
Nonfibrous material	3.6	2611
pH	3.7	2811
Shrinkage	3.8	5550
Seam efficiency	3.9	5110 6/
Flame resistance:		
Initial	3.12	5903
After 15 launderings	3.12	4.4.1
Toxicity	3.12.2	Comm. Std. 1/
Tearing strength (treated)	3.12.1	5132

1/ Unless otherwise specified, a supplier's certificate of compliance is required and will be acceptable for the stated requirement.

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2/ One determination shall be made from each sample unit and the result reported as "pass" or "fail".
3/ Method 5660 shall be used, except that exposure of the specimen and standard sample shall be for 20 standard hours. Any specimen showing discoloration less than or equal to that of the standard sample shall be rated "pass". Any specimen showing discoloration greater than that of the standard sample shall be rated "fail". When no standard sample is available, the discoloration shall be considered satisfactory if it is not worse than the rating of "fair" indicated in Method 5660.
4/ The specimens must be dried after each of the 3 laundering cycles.
5/ On the color transfer cloth evaluation, only the stain on the cotton fibers of the color transfer cloth shall be evaluated.
6/ The needle shall be 0.032 ± 0.001 inch across the blade at the eye and shall have a round point. The thread shall be type IA1 conforming to V-T-276; the needle thread shall be ticket No. 60, 3 ply and the looper thread shall be ticket No. 70, 2 ply.

4.4.1 Durability of treatment to laundering. The laundering shall be performed in accordance with Method 5556 (cotton laundering procedure), except that the load shall consist of twenty-four 18-inch squares of material and sufficient ballast in similar size pieces to effect the required 20-pound load. The specimens for the flame-resistance tests shall be conditioned for a minimum of 16 hours at standard atmospheric conditions for textiles prior to testing.

5. PREPARATION FOR DELIVERY

5.1 Put-up and packaging. Put-up and packaging shall be level A, B or C as specified (see 6.2).

5.1.1 Levels A and C. The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1134.

6. NOTES

6.1 Intended use. The cloth covered by this specification is intended specifically for men's and women's shirts, nurses caps, and pajamas.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of cloth of this specification.
(b) Types and classes of cloth required (see 1.2.1).
(c) Color of cloth required (see 3.3).
(d) Width of cloth required (see 3.4.1).
(e) When length of piece is other than specified (see 3.10).
(f) Whether flame-resistant treated cloth is required (see 3.12).
(g) Selection of applicable levels of put-up, packaging and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the procuring office issuing the invitation for bids.

6.4 Commercial grey designation. It is suggested that cloth of the following commercial grey designations (or prorata) be used to produce cloth meeting the requirements of this specification:

Commercial grey designation (suggested)

Type	Width	Yarns per pound	Yards per pound
I	45	144 x 76	3.46
II	47	136 x 60	3.20
III	41	98 x 56	3.80

6.5 Flame-resistant cloth. When Military agencies require flame-resistant cloth, the requirements of 3.12 must first be approved by the U.S. Army Natick Laboratories.

6.6 Sears Fabric Defect Scales are available from Sears, Roebuck and Company, Department 769, "Fabric Defect Replica Kit", 925 South Homan Avenue, Chicago, IL 60607.

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Navy - SA
Air Force - 11

Review activities:

Army - MD
Navy - MC
Air Force - 45

User activity:

Army - ME

Preparing activity:

Army - GL

Civil Agency Coordinating Activities:

GSA-FSS
HEW-NIH
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

CCC-C-437D CLOTH, BROADCLOTH, COTTON MERCERIZED

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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